Taha Afzal

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WORK EXPERIENCE

Software Engineer II, CNH Dec 2024 – Present | Sioux Falls, SD

- Designing and implementing features connecting user-facing functionality with vehicle subsystems using Qt and C++ in an Agile team.
- Developing robust C++ interfaces that integrate the UI with underlying system components and enable reliable system functionality.
- Building embedded Linux images with Yocto, writing BitBake recipes and layers to package applications for target hardware.
- Programming vehicle ECUs by discovering controllers on the network with mDNS/Avahi service and flashing them over Apache Thrift RPC.
- Debugging multi-threaded software using gdb-multiarch, analyzing system logs, and documenting code with Doxygen.
- Improving release cycles by 20% by implementing and maintaining CI/CD pipelines, automating build, test, and release workflows.

Software Engineer, CNH May 2022 – Nov 2024 | Sioux Falls, SD

- Collaborated with the team to transition the Viper 4+ build system from qmake to CMake, reducing build times by 25%.
- Implemented features to create field boundaries, ensuring efficient chemical usage and preventing accidental sprays in waterways.
- Enhanced a file conversion library, expanding support to JSON, CSV, and XML formats, ensuring 99.9% error-free conversions.
- Mentored an intern in implementing the Delta-T quality measurement system to assess spray effectiveness based on weather conditions.

Software Engineer Intern, Raven Industries, Inc. Sep 2021 – May 2022 | Sioux Falls, SD

- Implemented on-device creation of geospatial vector data format shapefiles for the multiline feature in Raven Operating Software.
- Developed this highly-requested feature for the Field Computer product line using Qt and C++ in an Agile environment.
- Improved the multiline feature to make it more user-friendly and easier to discover enabling farmers to save 60% of their time.

Software Engineer Intern, Raven Industries, Inc. May 2021 – Aug 2021 | Sioux Falls, SD

- Collaborated in an Agile team of 8 to implement single sign-on for all 4 customer-facing apps for a secure user-authorization process.
- Undertook displaying telemetry and generating alerts with Grafana to monitor load and throughput of critical services more closely.
- Streamlined infrastructure with Terraform, reduced moving parts, refactored code, and improved documentation.

SKILLS

C++ | Python | Rust | Swift | JavaScript | SQL | Bash | ReactJS | Qt | QML | Terraform | AWS | Docker | Git | Java

EDUCATION

M.S. Computer Science, Georgia Institute of Technology Jan 2026 – Present

Relevant Coursework: Advanced Operating Systems, High Performance Computer Architecture, Computer Networks

B.A. Computer Science and Software Engineering, Augustana University Aug 2018 – May 2022

Relevant Coursework: Advanced Data Structures, C++ Programming, Comp Organization & Architecture, Operating Systems, Advanced Programming

Honors and Awards

Global Leader Scholar, Slate Ronning International Scholar, Fred C. and J. Marie Christopherson Scholar, AURAFA Award Recipient

PROJECTS

Uni Go, Co-Founder Oct 2023 – Present

- Co-founded the re-branded development of Ridesio to bring an affordable, and communal mode of transportation to college campuses.
- Partnering with 3 college campuses to integrate Uni Go, delivering a tailored solution that meets the unique needs of each college.

Health Memo, Tech Lead Oct 2020 – Nov 2020

- Led a team of 3 to develop an iOS app to help the elderly manage their chronic diseases and share regular updates with their doctor.
- Utilized EMRs and Google's Text-to-Speech and Speech-to-Text to create a personalized experience for patients.
- Won $9,000 along with an invitation to work with engineers at Sanford Health for further research and development of the project.

LEADERSHIP AND AFFILIATIONS

Augustana University, Trustee Dec 2024 – Present

- Serving as a trustee on the Board of Trustees at Augustana, focusing on the well-being of faculty, students, and staff.
- Strengthening the Computer Science program, advancing international education, and driving entrepreneurial initiatives on campus.